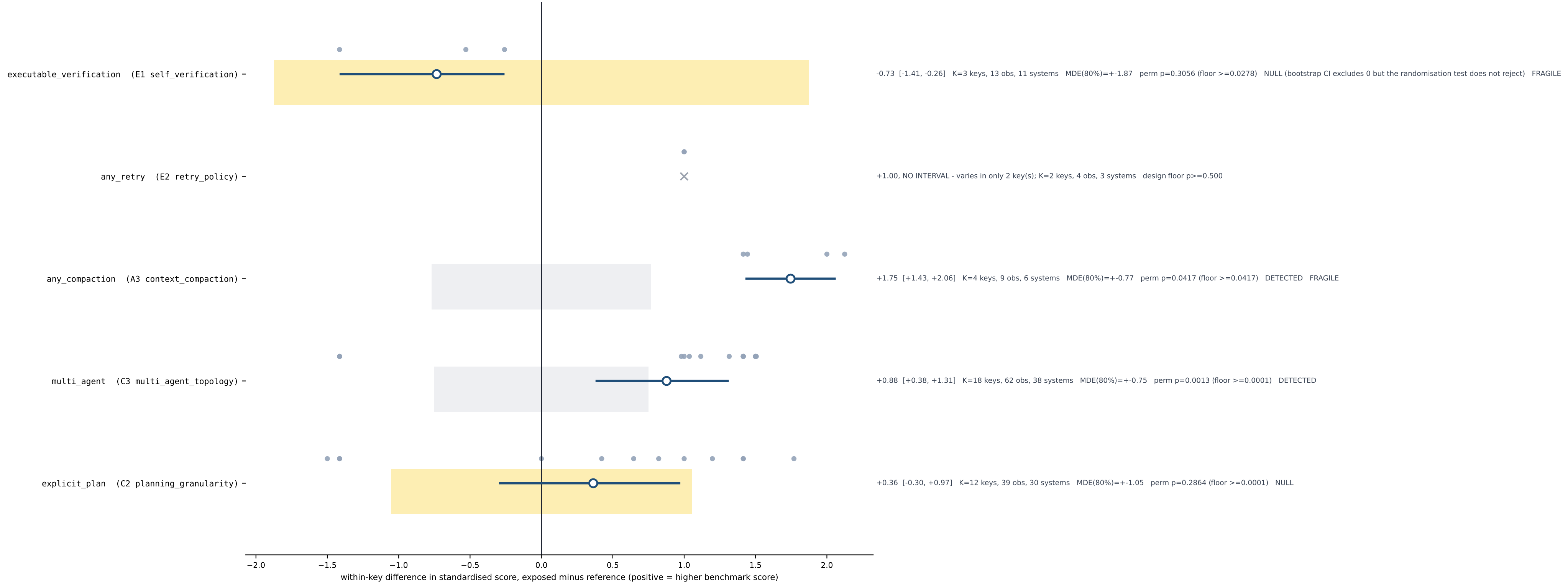


Five pre-specified harness-design contrasts, estimated within benchmark x split x model
Open circle: mean of key-level differences. Bar: percentile bootstrap CI resampling KEYS (not rows). Shaded band: +- the minimum detectable effect at 80% power (amber = only a large effect was detectable).
Small grey dots above each row: the individual key-level differences the estimate is made of - the evidence, of which the interval is only a summary.



All five contrasts were fixed as a set before estimation and all five are reported, including the null ones and the ones the data cannot support; there is no stepwise selection and nothing was dropped for its result. A key contributes only if the contrast varies inside it, which is what makes each estimate a within-key one. A CI covering zero means no detectable association at this sample size - read it together with the minimum detectable effect on the same line, which is what could have been found. A row marked NO INTERVAL is one where the design cannot support inference at all. Scaffold drift (docs/benchmark_caveats.md sec. 5): a leaderboard or paper row does not say which commit of the harness produced its number, while HARNESS-DB codes each system at a pinned commit, so the harness coded may not be the harness that scored. That is measurement error on the design variable, not the outcome; no fixed effect absorbs it and it attenuates estimates towards zero. Every effect shown is a LOWER BOUND on magnitude. Reporting is also self-selected (sec. 3): authors report where their system does well, so a within-key comparison is not a randomised one. Associational only; no causal reading is supported.

Power, computed from the observed between-key spread and from the design itself:

- executable_verification: with 3 keys, 13 observations and 11 distinct systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 1.87 within-key sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. FRAGILE: fewer than 5 keys, so the between-key spread that both the interval and this number rest on is itself barely estimated; treat the interval as descriptive. The design admits 36 distinct within-key label assignments, so the smallest attainable permutation p-value is 0.0278 (observed 0.3056).
- any_retry: NOT ESTIMABLE - the contrast varies inside only 2 comparable key(s), so no interval and no minimum detectable effect are defined. The design admits 4 distinct within-key label assignments, so the smallest attainable permutation p-value is 0.5000; no conventional significance level is reachable from this contrast whatever the point estimate.
- any_compaction: with 4 keys, 9 observations and 6 distinct systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 0.77 within-key sd. Effects smaller than that are beyond the resolution of this dataset. FRAGILE: fewer than 5 keys, so the between-key spread that both the interval and this number rest on is itself barely estimated; treat the interval as descriptive. The design admits 24 distinct within-key label assignments, so the smallest attainable permutation p-value is 0.0417 (observed 0.0417).
- multi_agent: with 18 keys, 62 observations and 38 distinct systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 0.75 within-key sd. Effects smaller than that are beyond the resolution of this dataset. The design admits 134,120,448,000 distinct within-key label assignments, so the smallest attainable permutation p-value is 0.0001 (observed 0.0013).
- explicit_plan: with 12 keys, 39 observations and 30 distinct systems, the smallest effect detectable at 80% power (two-sided, alpha=0.05) is 1.05 within-key sd. Only a very large effect would have been detectable here, so the absence of a detected association is uninformative about anything smaller. The design admits 622,080 distinct within-key label assignments, so the smallest attainable permutation p-value is 0.0001 (observed 0.2864).